

A UNIFORMLY ATOMIC TYPE-I SUBCASE OF A STABILITY PROBLEM FOR NONCOMMUTATIVE SYMMETRIC SPACES

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STATUS AND SOURCE

This packet gives a candidate partial positive answer to Problem 9 in Czerwinska–Kaminska [1], arXiv:1704.02033, PDF page 55. The source asks:

“Assume E is stable and $\mathcal{M}$ is of type I. Is the space $E(\mathcal{M}, \tau)$ stable?”

The result below answers this affirmatively for countable atomic type-I algebras whose positive factor trace weights are uniformly bounded below. The weights need not be commensurable. It does not settle the diffuse-center case or atomic weights tending to zero.

The original page also records the Arazy–Raynaud theorem used below and is shown here at full readable width.

summary complete.
The main result lifting the property of stability from E to C_E was obtained independently by J. Arazy [7] and Y. Raynaud [92].

Theorem 20.5. *Let E be a symmetric separable sequence space. Then E is stable if and only if C_E is stable.*

Here we use the fact that E is an isometric subspace of C_E (see Proposition 3.1) and clearly stability is inherited by subspaces, so if C_E is stable then E is stable. The non-trivial proof is in the opposite direction. The proof is based on Propositions 20.2 20.3 Corollary 20.4 and two technical lemmas.

In 1997 Marcolino Nhany found a necessary and sufficient condition for stability of noncommutative $L_p(\mathcal{M}, \tau)$ spaces.

Theorem 20.6. [82] *La Théorème principale] The following properties are equivalent.*

- (1) *The von Neumann algebra $\mathcal{M}$ is of type I.*
- (2) *$L_p(\mathcal{M}, \tau)$ is stable for all $1 \leq p < \infty$.*
- (3) *There exists $1 \leq p < \infty$, $p \neq 2$, such that $L_p(\mathcal{M}, \tau)$ is stable.*

Remark 20.7. The semifinite von Neumann algebra $\mathcal{M}$ is always of type I or type II. For precise definition of types of von Neumann algebra we refer to [103].

Problem 9. Assume E is stable and $\mathcal{M}$ is of type I. Is the space $E(\mathcal{M}, \tau)$ stable?

DEFINITIONS

A metric space (X, d) is *stable* if for every two bounded sequences (x_m) and (y_n) and every two free ultrafilters $\mathcal{U}, \mathcal{V}$ on $\mathbb{N}$,

$$\lim_{m, \mathcal{U}} \lim_{n, \mathcal{V}} d(x_m, y_n) = \lim_{n, \mathcal{V}} \lim_{m, \mathcal{U}} d(x_m, y_n).$$

A Banach space is stable when its norm metric is stable. Stability is inherited by isometric subspaces.

Date: 2026-07-19.

For a symmetric Banach sequence space F , its unitary matrix ideal is

$$C_F = \{T \in \mathcal{K}(K) : s(T) = (s_j(T))_{j \geq 1} \in F\}, \quad \|T\|_{C_F} = \|s(T)\|_F,$$

where $s_j(T)$ are the singular values in decreasing order. The Arazy–Raynaud theorem says that a separable symmetric sequence space F is stable if and only if C_F is stable [2, 1].

For a semifinite von Neumann algebra $(\mathcal{M}, \tau)$ and a symmetric Banach function space E , the noncommutative symmetric space $E(\mathcal{M}, \tau)$ consists of the τ -measurable operators x for which the generalized singular-value function $\mu_\tau(x)$ belongs to E , with $\|x\|_{E(\mathcal{M}, \tau)} = \|\mu_\tau(x)\|_E$.

PROOF INTUITION

The obstruction in the full problem is that a general type-I trace need not look like ordinary counting measure on one Hilbert space. In the atomic commensurable case it does, after an exact amplification. If the trace on the n th factor is $\delta k_n \text{Tr}$, tensor that factor with the identity on $\mathbb{C}^{k_n}$. Every singular value is then repeated k_n times, so the ordinary spectral counting function of the amplified operator is exactly $1/\delta$ times its original τ -distribution function.

Equal intervals of length δ cut a symmetric sequence space F_δ out of E . The amplification identifies $E(\mathcal{M}, \tau)$ isometrically with a closed subspace of C_{F_δ} . Because F_δ is a closed subspace of the stable space E , it is stable; Arazy–Raynaud then makes C_{F_δ} stable. If the original weights merely have a positive lower bound, round them upward to multiples of a common, very small δ . This changes every norm by a factor at most $1 + \varepsilon$. Stability survives the limit of these arbitrarily small distortions. Diffuse centers and weights tending to zero remain open after this argument.

PARTIAL THEOREM

Theorem 1 (Commensurable atomic case). *Let E be a stable, separable symmetric Banach function space on $[0, \infty)$ with order-continuous norm. Let*

$$\mathcal{M} = \bigoplus_{n \geq 1}^{\ell_\infty} B(H_n),$$

where every H_n is a separable Hilbert space. Suppose that, for some $\delta > 0$ and positive integers k_n , the faithful normal semifinite trace is

$$\tau((x_n)) = \delta \sum_{n \geq 1} k_n \operatorname{Tr}_{H_n}(x_n), \quad (x_n) \in \mathcal{M}_+.$$

Then $E(\mathcal{M}, \tau)$ is stable.

Proof. Put $I_j = [(j-1)\delta, j\delta)$ and let X_δ be the closed linear span in E of the pairwise disjoint indicators $\mathbf{1}_{I_j}$. Define the coefficient sequence space

$$F_\delta = \left\{ a = (a_j) : \sum_{j \geq 1} a_j \mathbf{1}_{I_j} \text{ converges in } E \right\}, \quad \|a\|_{F_\delta} = \left\| \sum_{j \geq 1} a_j \mathbf{1}_{I_j} \right\|_E.$$

Because the intervals are equimeasurable and E is symmetric, F_δ is a symmetric Banach sequence space. Its step-function map is a linear isometry onto X_δ . Thus F_δ is stable, since X_δ is a closed subspace of the stable Banach space E ; it is also separable. The Arazy–Raynaud theorem therefore implies that C_{F_δ} is stable.

Let

$$K = \bigoplus_{n \geq 1} (H_n \otimes \mathbb{C}^{k_n}) \quad \text{and} \quad J((x_n)) = \bigoplus_{n \geq 1} (x_n \otimes I_{k_n}).$$

Functional calculus commutes with this amplification. Consequently, for every x affiliated with $\mathcal{M}$ and every $\lambda \geq 0$,

$$\begin{aligned} \operatorname{Tr}_K(\mathbf{1}_{(\lambda, \infty)}(|Jx|)) &= \sum_{n \geq 1} k_n \operatorname{Tr}_{H_n}(\mathbf{1}_{(\lambda, \infty)}(|x_n|)) \\ &= \delta^{-1} \tau(\mathbf{1}_{(\lambda, \infty)}(|x|)). \end{aligned}$$

The equality is valid in $[0, \infty]$. Taking generalized inverses of these distribution functions gives

$$\mu_\tau(x)(t) = \mu_{\operatorname{Tr}_K}(Jx)(t/\delta), \quad t \geq 0.$$

Set

$$a_j = \mu_{\operatorname{Tr}_K}(Jx)(j-1), \quad j \geq 1.$$

These are the decreasing spectral values of the amplification (with extended values allowed at this point). The scaling identity says exactly

$$(1) \quad \mu_\tau(x) = \sum_{j \geq 1} a_j \mathbf{1}_{I_j}.$$

Now take $x \in E(\mathcal{M}, \tau)$. Order continuity of E makes the finite truncations of the right side of (1) converge to $\mu_\tau(x)$ in E . Hence $a = (a_j) \in F_\delta$. Since a separable symmetric sequence space is contained in c_0 , the sequence tends to zero. Thus Jx is a compact bounded operator, $a = s(Jx)$ is its usual singular-value sequence, and $Jx \in C_{F_\delta}$. Moreover,

$$\|x\|_{E(\mathcal{M}, \tau)} = \|\mu_\tau(x)\|_E = \|s(Jx)\|_{F_\delta} = \|Jx\|_{C_{F_\delta}}.$$

Thus J is a linear isometry from $E(\mathcal{M}, \tau)$ into C_{F_δ} . Its range is closed because the domain is complete. Stability of C_{F_δ} passes to this closed subspace, proving the claim. $\square$

Lemma 1 (Almost-isometric stability test). *Let X be a Banach space. Suppose that for every $c > 1$ there are a stable Banach space Y_c and a linear map $T_c : X \rightarrow Y_c$ such that*

$$\|z\|_X \leq \|T_c z\|_{Y_c} \leq c\|z\|_X \quad (z \in X).$$

Then X is stable.

Proof. For bounded sequences $(x_n), (y_m)$ and ultrafilters $\mathcal{U}, \mathcal{V}$, let A and B be the two iterated limits of $\|x_n + y_m\|_X$ in opposite orders. Stability of Y_c says that the corresponding two iterated limits of $\|T_c x_n + T_c y_m\|_{Y_c}$ have one common value L_c . The distortion bounds give

$$A \leq L_c \leq cA, \quad B \leq L_c \leq cB.$$

Hence $A \leq cB$ and $B \leq cA$ for every $c > 1$. Letting $c \downarrow 1$ gives $A = B$. $\square$

Theorem 2 (Uniformly atomic case). *Let E be as in Theorem 1, and let*

$$\mathcal{M} = \bigoplus_{n \geq 1}^{\ell_\infty} B(H_n), \quad \tau((x_n)) = \sum_{n \geq 1} w_n \operatorname{Tr}_{H_n}(x_n),$$

where $w_n > 0$ and $w_ := \inf_n w_n > 0$. Then $E(\mathcal{M}, \tau)$ is stable. No commensurability assumption on (w_n) is required.*

Proof. Fix $c > 1$. Choose $\delta > 0$ with $\delta < (c-1)w_*$, put $k_n = \lceil w_n/\delta \rceil$, and set $v_n = \delta k_n$. Then

$$w_n \leq v_n < cw_n \quad (n \geq 1).$$

Let $\tau_v = \sum_n v_n \operatorname{Tr}_{H_n}$. For every affiliated x and every $\lambda \geq 0$, the two distribution functions satisfy

$$d_\tau(\lambda; x) \leq d_{\tau_v}(\lambda; x) \leq c d_\tau(\lambda; x).$$

Taking generalized inverses yields

$$\mu_\tau(x)(t) \leq \mu_{\tau_v}(x)(t) \leq \mu_\tau(x)(t/c) =: D_c \mu_\tau(x)(t).$$

The stable symmetric space E is order continuous and hence fully symmetric in the source framework. For decreasing $f \geq 0$,

$$\int_0^t D_c f(s) ds = c \int_0^{t/c} f(u) du \leq c \int_0^t f(u) du,$$

so $D_cf \prec\prec cf$ and $\|D_cf\|_E \leq c\|f\|_E$. Consequently

$$\|x\|_{E(\mathcal{M},\tau)} \leq \|x\|_{E(\mathcal{M},\tau_v)} \leq c\|x\|_{E(\mathcal{M},\tau)}.$$

The identity is therefore a distortion- c linear embedding of $E(\mathcal{M},\tau)$ into $E(\mathcal{M},\tau_v)$. The latter is stable by Theorem 1. Lemma 1, applied for every $c > 1$, proves the claim. $\square$

VERIFIER REPORT

The proof was checked independently along the following chain.

- (1) The equal-step coefficient space is not treated as an arbitrary set of step functions: it is defined through the closed span of the equal interval indicators. This makes completeness and the isometric embedding into E explicit.
- (2) At a spectral threshold λ , tensoring the n th block with I_{k_n} multiplies its spectral rank by exactly k_n . This proves the distribution identity before any norm argument and avoids an implicit reordering assumption.
- (3) Generalized inversion scales the distribution variable by δ , which yields intervals of length δ in (1); no missing factor remains in the norm equality.
- (4) Order continuity is used exactly once, to pass from finite singular-value steps to the full function in E . Separability of F_δ then guarantees compactness of the amplification.
- (5) A reusable finite check in `code/check_finite_distribution.py` tested six block families at nine thresholds and returned PASS. It checks only the elementary integer-multiplicity identity and is not part of the proof.
- (6) For the upgraded theorem, the inequalities between the two distribution functions were inverted explicitly, and the only dilation estimate used is $D_cf \prec\prec cf$, valid because stable E is order continuous and hence fully symmetric in the source framework.

No unproved mathematical dependency remains inside the stated theorem. The external dependency is the cited Arazy–Raynaud stability theorem for unitary matrix ideals. Recommended human-review focus: confirm the source convention under which stability of a symmetric function space supplies the stated separability/order-continuity setting, and recheck the generalized-inverse scaling in (1) and in the trace-perturbation argument of Theorem 2.

LIMITATIONS AND NOVELTY STATUS

The upgraded theorem does not address type-I algebras with diffuse center or a countable atomic direct sum whose positive trace weights tend to zero. In those cases no single commensurable trace gives a uniformly small relative

perturbation. Local or finite-dimensional approximation is insufficient by itself because stability is not merely a finite-representability property.

Cheap run indexes were searched for arXiv:1704.02033, the exact title, and the core stability/type-I/unitary-ideal terms. A bounded web search on 2026-07-19 using the exact question, atomic/commensurable variants, and uniformly positive trace weights found the source and the classical sequence-ideal theorem but no later resolution or duplicate of this trace-perturbation reduction. This is a bounded novelty check, not an exhaustive claim of priority. The status is therefore `candidate_partial_likely_valid`, pending human review.

REFERENCES

- [1] M. M. Czerwinska and A. Kaminska, *Geometric properties of noncommutative symmetric spaces of measurable operators and unitary matrix ideals*, arXiv:1704.02033 (2017), Problem 9, PDF p. 55.
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- [3] Y. Raynaud, *Stabilite des espaces C_E* , Seminaire de Geometrie des Espaces de Banach, Universite Paris 7, 1982–1983.